

CSA1017 – SOFTWARE ENGINEERING – DOCKER LAB

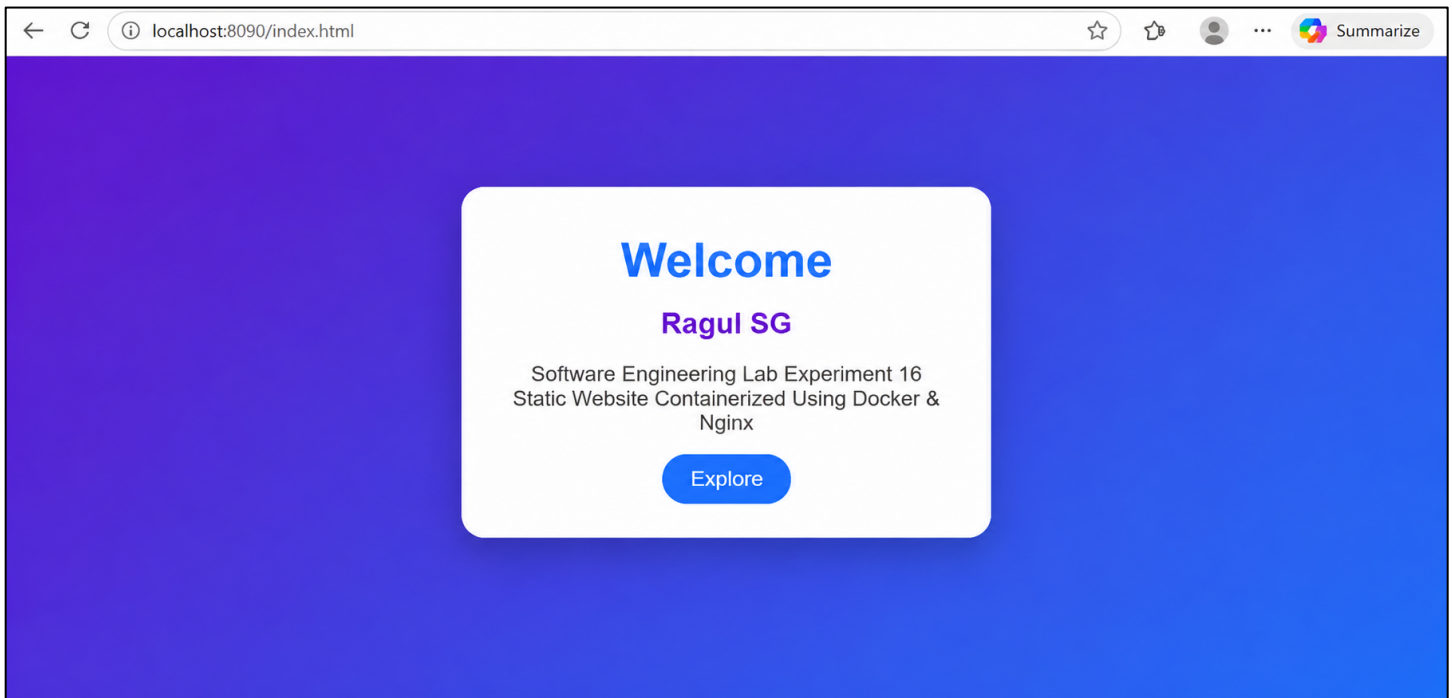
RAGUL SG - 192525007

EXPERIMENT-16

AIM:

To create a Static Website and Containerize, Build & Serve it using Docker. Tasks: Create a Simple Static Website (index.html file) with basic HTML content. Write/create a Dockerfile to serve the website using Nginx. Build the Docker Image. Run the container. Access the Website using a Browser.

OUTPUT:



```
C:\Users\HP 830 G7\OneDrive\Documents\R programming project\Desktop\my website>docker run -d -p 8081:80 mywebsite
fb1181dd47db5f4167bf5ee7fae0b326838de0d43cac77949e09972e713d41aa

What's next:
  Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint
t agitated_bohr (17c3a501a3af06cddb8f1bf23efe30e8e16110758c729f4cffbac6095b16237f): Bind for 0.0.0.0:8081 failed: port is already al
located

Run 'docker run --help' for more information

C:\Users\HP 830 G7\OneDrive\Documents\R programming project\Desktop\my website>docker run -d -p 8082:80 mywebsite
274e96af32553c9d600ed8b70e454f4de812564316a33ed78d808d20a32e839a

C:\Users\HP 830 G7\OneDrive\Documents\R programming project\Desktop\my website>docker run -d -p 8090:80 mywebsite
e760e96218fc3aafca90dbfb3f017db7169bd06f5fed2d77c0b32c1572fe6316

C:\Users\HP 830 G7\OneDrive\Documents\R programming project\Desktop\my website>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
NAMES
e760e96218fc  mywebsite     "/docker-entrypoint...." About a minute ago Up About a minute 0.0.0.0:8090->80/tcp, [::]:8090->80/
cp_sleepy_wing
274e96af3255  mywebsite     "/docker-entrypoint...." About a minute ago Up About a minute 0.0.0.0:8082->80/tcp, [::]:8082->80/
cp_awesome_sanderson
c34819c08be9  ce720e993ebe  "/docker-entrypoint...." 7 minutes ago  Up 7 minutes  0.0.0.0:8081->80/tcp, [::]:8081->80/
cp_serene_montalcini
d858af98e40a  nginx         "/docker-entrypoint...." 10 hours ago  Up 10 hours  0.0.0.0:8080->80/tcp, [::]:8080->80/
cp_my-nginx
a44c4df79c61  python:3.12   "python3"               12 hours ago  Up 12 hours
rpython

C:\Users\HP 830 G7\OneDrive\Documents\R programming project\Desktop\my website>
```

Gordon

Containers

Images

Volumes

Kubernetes

Builds

MCP Toolkit

Docker Hub

Logs

Models

Extensions

Images

Give feedback

Local

My Hub

0 Bytes / 0 Bytes in use

3 images

Last refresh: 12 hours ago

Search

		Name	Tag	Image ID	Created	Size	Actions
		python	3.12	ea7b35cdb10b	8 days ago	1.61 GB	
		nginx	latest	6415da96b72a	19 hours ago	240.55 MB	
		mywebsite	latest	863f92f35f76	8 minutes ago	237.72 MB	

Walkthroughs

```
C:\WINDOWS\system32\cmd. x + - □ x
C:\Users\HP 836 G7\OneDrive\Documents\R programming
project\desktop my website>docker run -d -p 8090:80.0
27be96da732038c9d688cdb87ae5eafdc6812364316a33ed7f
8d9602c8a32c839a

C:\Users\HP 836 G7\OneDrive\Documents\R programming
project\desktop my website>docker ps
CONTAINER ID    IMAGE    COMMAND
CREATED        STATUS    PORTS
NAMES
27be96da7320    mywebsite    "/docker-entrypoint..."
About a minute ago    Up About a minute
0.0.0.0:8090->80/tcp, [::]:8090->80/tcp
sleepy_wing

C:\Users\HP 836 G7\OneDrive\Documents\R programming
project\desktop my website>docker ps
CONTAINER ID    IMAGE    COMMAND
CREATED        STATUS    PORTS
NAMES
27be96da7320    mywebsite    "/docker-entrypoint..."
About a minute ago    Up About a minute
0.0.0.0:8090->80/tcp, [::]:8090->80/tcp
sleepy_wing

C:\Users\HP 836 G7\OneDrive\Documents\R programming
project\desktop my website>docker ps -a
CONTAINER ID    IMAGE    COMMAND
CREATED        STATUS    PORTS
NAMES
27be96da7320    mywebsite    "/docker-entrypoint..."
7 minutes ago    Up 7 minutes
0.0.0.0:8090->80/tcp, [::]:8090->80/tcp
sleepy_wing

C:\Users\HP 836 G7\OneDrive\Documents\R programming
project\desktop my website>_
```

My Static Website

localhost:8090/index.html

P

Welcome

Ragul SG

Software Engineering Lab Experiment 16
Static Website Containerized Using Docker &
Nginx

Explore

EXPERIMENT-22

AIM:

To Push and Pull Docker Images Using Docker Hub Tasks: Create a Docker image for a static HTML website. Tag and push the image to Docker Hub. Pull the image from Docker Hub and run it on another machine. Submit Docker commands and screenshots for each step.

OUTPUT:

```
C:\Users\HP 830 G7\OneDrive\Documents\R programming project\Desktop\my website>docker login
```

USING WEB-BASED LOGIN

 Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: HBVV-MTZQ

Press ENTER to open your browser or submit your device code here: <https://login.docker.com/activate>

Waiting for authentication in the browser...

Login Succeeded

```
C:\Users\HP 830 G7\OneDrive\Documents\R programming project\Desktop\my website>
```

```
C:\Users\HP 830 G7\OneDrive\Documents\R programming project\Desktop\my website>docker run -d -p 8081:80 mywebsite
34819c08be95164deb9ca142433bb488a70952ff7be51cff52a5bbbad8ad8b2
```

```
C:\Users\HP 830 G7\OneDrive\Documents\R programming project\Desktop\my website>docker build -t mywebsite .
```

```
[+] Building 2.1s (7/7) FINISHED                                docker:desk
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 104B
=> [internal] load metadata for docker.io/library/nginx:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 1.68kB
=> CACHED [1/2] FROM docker.io/library/nginx:latest@sha256:6415da96b72a2f6ff433053df1da4bd507e45bc44b2c83dbac0ddeea80c6f06
=> => resolve docker.io/library/nginx:latest@sha256:6415da96b72a2f6ff433053df1da4bd507e45bc44b2c83dbac0ddeea80c6f06
=> [2/2] COPY index.html /usr/share/nginx/html/index.html
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:8fae4049132a6d3a72bc9d9364f9affa0e9d70a73bcd38e62eff59a7a74fc263
=> => exporting config sha256:29fbae6e621a56cecbef47039a959de1a1947576c6fa1337cced9b464a348b96
=> => exporting attestation manifest sha256:28d07a357b3768bafda07efd7ef99dc107b6f15096c0db87c16ab4640ccea3f7
=> => exporting manifest list sha256:863f92f35f76f377e41dbd600f38e3466aa2ea05e3021f415f5dd108a4cd6c31
=> => naming to docker.io/library/mywebsite:latest
=> => unpacking to docker.io/library/mywebsite:latest
```

```
C:\Users\HP 830 G7\OneDrive\Documents\R programming project\Desktop\my website>docker run -d -p 8081:80 mywebsite
b1181dd47db5f4167bf5ee7fae0b326838de0d43cac77949e09972e713d41aa
```

```
C:\Users\HP 836 G7\OneDrive\Documents\R programming project\Desktop\my website>docker tag mywebsite:latest bhindya/mywebsite:v1
C:\Users\HP 836 G7\OneDrive\Documents\R programming project\Desktop\my website>docker push bhindya/mywebsite:v1
The push refers to repository [docker.io/bhindya/mywebsite]
4c4a81c6b5b0: Pushed
991311f0c030: Mounted from library/nginx
7ae2f3151bcb: Mounted from library/nginx
b8d18c6b4ee8: Mounted from library/nginx
c2e83b6d4f69: Mounted from library/nginx
3d4b98acb725: Pushed
31610b29661b: Mounted from library/nginx
792e32108716: Mounted from library/nginx
b4aa01e35daa: Mounted from library/nginx
v1: digest: ea2d3c853f72529f9e8197eb1ddb60f5de3506aa2ea8b5e922f43b5fddd168a size: 856

C:\Users\HP 836 G7\OneDrive\Documents\R programming project\Desktop\my website>docker pull bhindya/mywebsite:v1
v1: Pulling from bhindya/mywebsite
Digest: ea2d3c853f72529f9e8197eb1ddb60f5de3506aa2ea8b5e922f43b5fddd168a
Status: Image is up to date for bhindya/mywebsite:v1
docker.io/bhindya/mywebsite:v1

What's next:
View a summary of image vulnerabilities and recommendations -> docker scout quickview bhindya/mywebsite:v1

C:\Users\HP 836 G7\OneDrive\Documents\R programming project\Desktop\my website>docker run -d -p 8090:80 bhindya/mywebsite:v1
e9d26b8ee38de140bb6f9b9dc85bf26bf83564b3af6b599d4612c4fc4813ce9b68

What's next:
Debug this container error with Garden -> docker ai "help me fix this container error"
```

Builds
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desktop-linux
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[Build history](#)
[Active builds](#)

☒ Show only my builds

☐	Name	ID	Builder	Durati...	Created	Cloud Bu...	Actions
☐	✓ my website	wg77ko	desktop-li...	1.7s	29 minutes ago	N/A	
☐	✓ my website	zv7v0x	desktop-li...	0.9s	34 minutes ago	N/A	
☐	✓ my website	vwesri	desktop-li...	2.7s	39 minutes ago	N/A	
☐	⚠ HP 830 G7	ioqxoc	desktop-li...	0.3s	55 minutes ago	N/A	

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Ragul SG
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Repositories

All repositories of Ragul SG are listed here.

All repositories
 ▼

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Name	Last updated ^	Contents	Visibility	Stars
mywebsite Static website containerized with Docker & Nginx	15 minutes ago	Public	Public	☆ 0

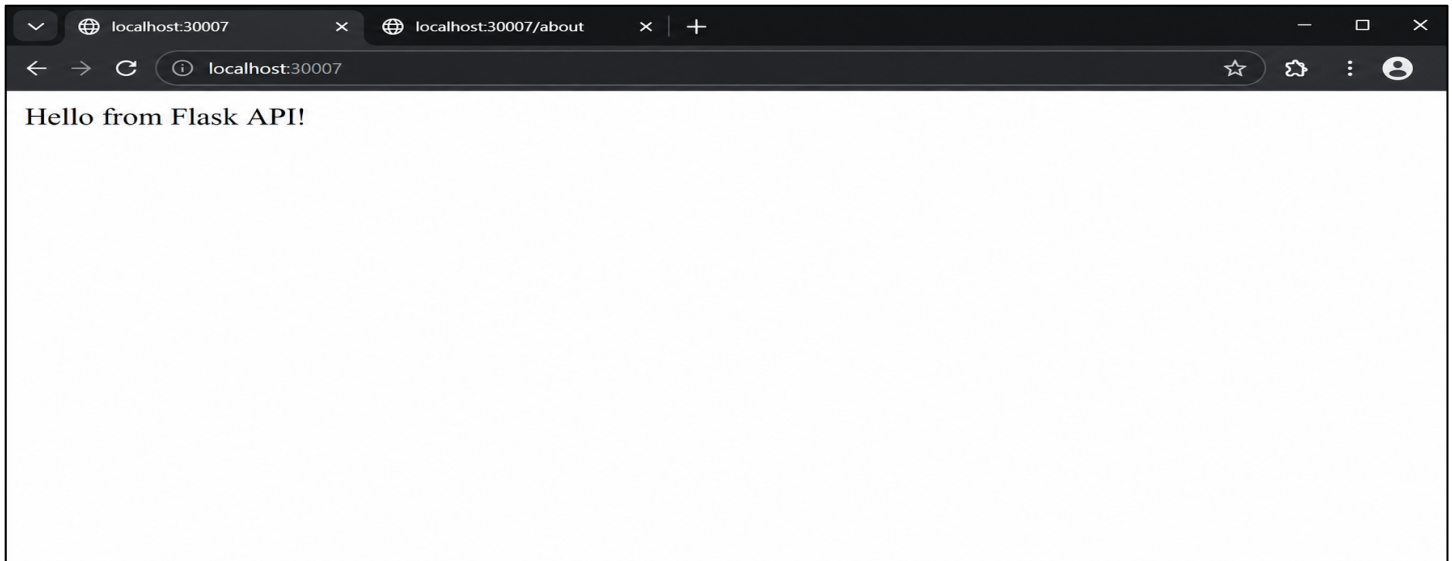
1-1 of 1
 <
>

EXPERIMENT – 17

AIM:

Create a Simple Python Flask API, Containerize the Application, Build & Push the Image using Docker and Deploy the Application using Kubernetes.

OUTPUT:

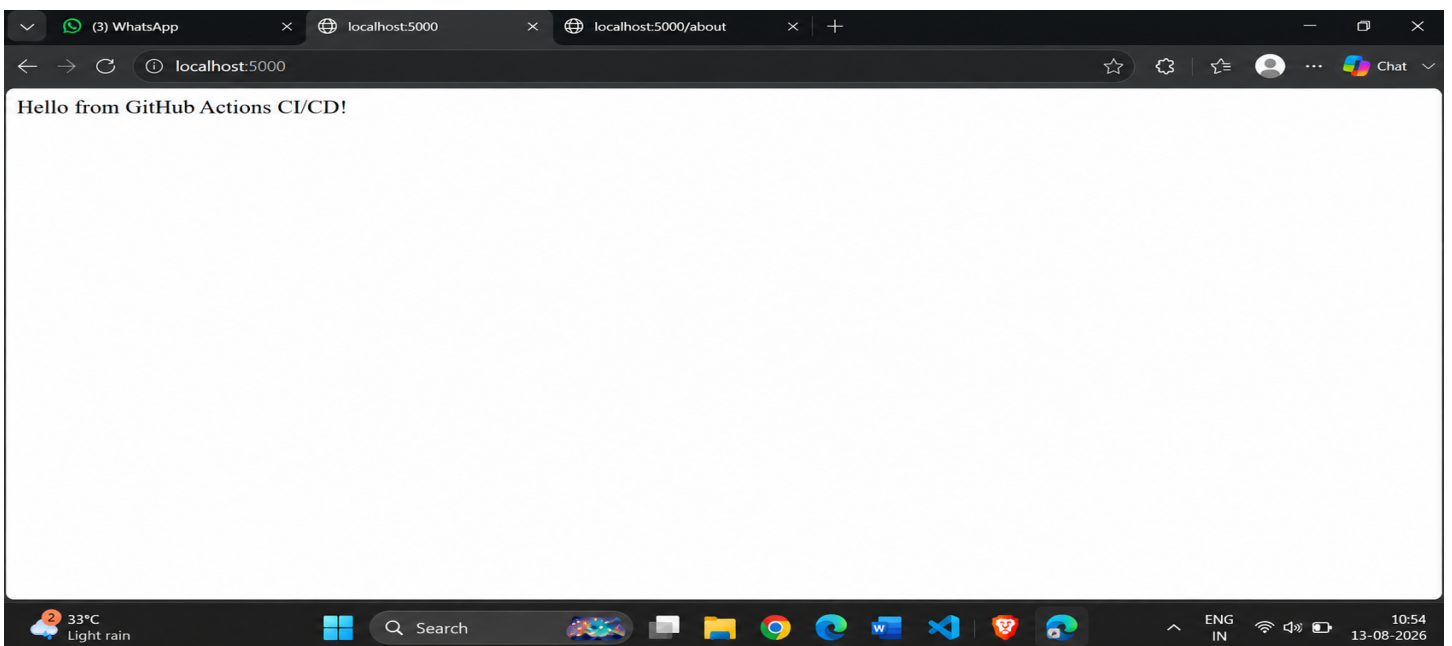


EXPERIMENT- 19

AIM:

Implement Continuous Deployment using GitHub Actions to deploy a Dockerized application.

OUTPUT:

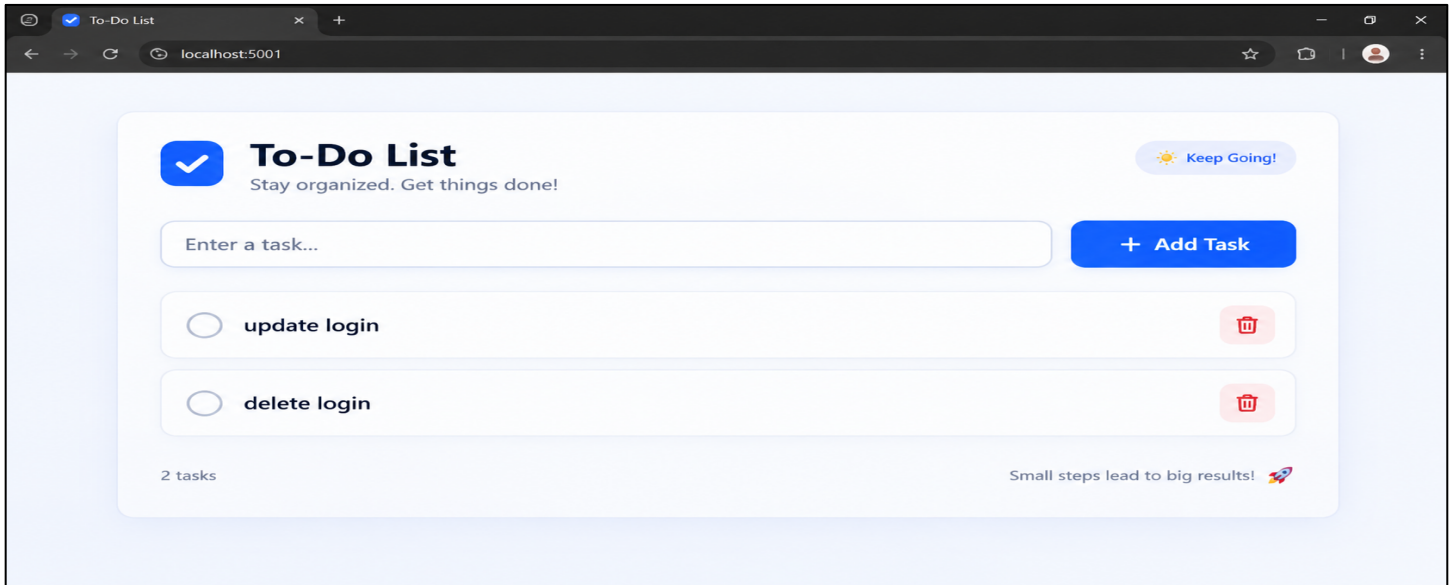


EXPERIMENT- 21

AIM:

Containerize a Python Flask Application Using Docker

OUTPUT:



EXPERIMENT- 23

AIM:

Deploy a Multi-Container Application Using Kubernetes

OUTPUT:

